Abderrahim Mechache – PhD Candidate in Computer Science
Lyon, France
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Academic Profile — LinkedIn

Education

PhD in Computer Science Jan 2024 – Present
LIRIS Laboratory, UMR 5205 CNRS, Claude Bernard University Lyon 1, France
Thesis: Modeling and analysis of complex 3D structures

Master's Degree in Computer Science (Rank: 2/38) 2021 – 2023
University of Abderrahmane MIRA, Béjaïa, Algeria
Major: Advanced Information Systems (SIA)

Bachelor's Degree in Computer Science (Rank: 2/72) 2019 – 2021
University of Abderrahmane MIRA, Béjaïa, Algeria
Major: Information Systems (SI)

Projects and Experience

PhD Project – GRADIENT [1, 2] Jan 2024 – Present
Graphs and Algorithms for 3D Protein Structure and Dynamics Classification
gradient.projet.liris.cnrs.fr

Teaching Assistant Sep 2025 – Dec 2025
Department of Computer Science, University of Lyon 1, France
Practical sessions for the course *"Databases and Web Technologies"* (Bachelor 2nd year)

Decision-Making in Autonomous Vehicles [3] Jan 2023 – Jun 2023
Master's Thesis Project — Research Track, University of Béjaïa, Algeria
Tools: Python, PyTorch, Pygame

SIAD – Information System for Decision Support Oct 2022 – Feb 2023
Master Project, University of Béjaïa, Algeria
Tools: UML, HTML, CSS, JS, VueJS, Python, Django, Scikit-learn

Skills

Research Domains: Deep Learning, Graph Neural Networks, Computer Vision, Graph Theory.
Programming: Python, PyTorch, PyTorch Geometric, TensorFlow, SQL, R, Java, Scikit-learn.
Languages: Kabyle (Native), Arabic (Fluent), French (Fluent), English (Intermediate).

References

[1] Abderrahim Mechache and Hamamache Kheddouci. Edge-enriched mesh representation for protein surface classification. In *Proceedings of the 37th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*. IEEE, 2025. Accepted for publication.

[2] Abderrahim Mechache and Hamamache Kheddouci. Enhancing protein classification with graph convolutional neural networks. In *International Conference on Pattern Recognition (ICPR)*, pages 109–124. Springer, 2024.

[3] Abderrahim Mechache, Tinhinane Chenache, Sofiane Aissani, and Mawloud Omar. Enhancing decision-making in autonomous vehicles: A study on the integration of k-nearest neighbor algorithm and reinforcement learning. In *Colloque sur les Objets et Systèmes Connectés*, 2023.